

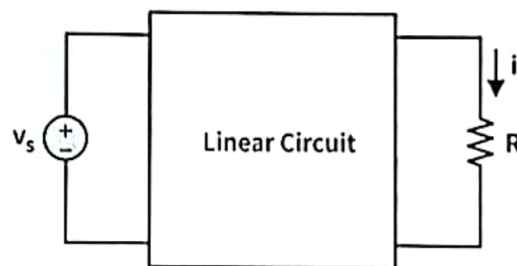
Linearity Property

- Linearity is the property of an element describing a linear relationship between cause and effect.
- It is a combination of the homogeneity property and the additivity property.
- *Homogeneity property* requires that if the input (excitation) is multiplied by a constant, then the output (response) is multiplied by the same constant.
- For a resistor, for example, Ohm's law relates the input i to the output v : $v=iR$. If i is increased by a constant k , then v increases correspondingly by k ; that is

$$k i R = k v$$

- *Additivity property* requires that the response to a sum of inputs is the sum of the responses to each input applied separately. Thus for a resistor, if $v_1 = i_1 R$ and, then applying $i_1 + i_2$ gives $v = (i_1 + i_2)R = i_1 R + i_2 R = v_1 + v_2$
- We say that a resistor is a *linear element* because the voltage-current relationship satisfies both the homogeneity and the additivity properties.
- In general, a circuit is linear if it is both additive and homogeneous.
- A *linear circuit* is one whose output is linearly related (or directly proportional) to its input. A linear circuit consists of only linear elements, linear dependent sources, and independent sources.

To illustrate the linearity principle, consider the linear circuit:



- The circuit has no independent sources inside it, is excited by a voltage source v_s (input), and is terminated by a load R . The current i through R can be taken as the output.
- Suppose $v_s = 10\text{ V}$ gives $i = 2\text{ A}$. According to the linearity principle, $v_s = 1\text{ V}$ will give $i = 0.2\text{ A}$. By the same token, $i = 1\text{ mA}$ must be due to $v_s = 5\text{ mV}$.
- When current i_1 flows through R , the power is $p_1 = R i_1^2$; when current i_2 flows through R , the power is $p_2 = R i_2^2$. If current $i_1 + i_2$ flows through R , the power absorbed is $p_3 = R(i_1 + i_2)^2 \neq p_1 + p_2$. Thus, the relationship between power and voltage (or current) is **nonlinear**.

Superposition Principle

- The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.
- To apply the superposition principle, we must keep two things in mind:
 1. Consider one independent source at a time while the others are turned off. Replace every voltage source by 0 V (short circuit), and every current source by 0 A (open circuit) to obtain a simpler circuit.
 2. Dependent sources are left intact as they are controlled by circuit variables.

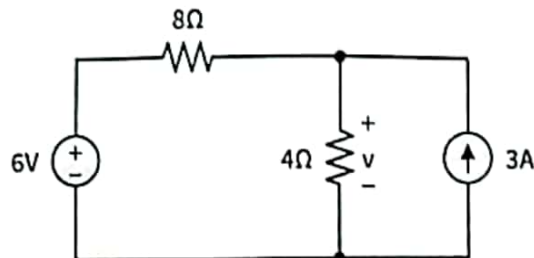
With these in mind, we apply the superposition principle in three steps:

Step 1: Turn off all independent sources except one source. Find the output (voltage or current) due to that active source.

Step 2: Repeat step 1 for each of the other independent sources.

Step 3: Find the total contribution by adding algebraically all the contributions due to the independent sources.

Using the superposition theorem to find v in the circuit below,

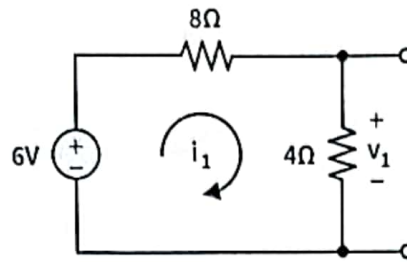


since there are two sources,

$$v = v_1 + v_2$$

where v_1 and v_2 are the contributions due to the 6-V voltage source and the 3-A current source, respectively.

To obtain v_1 , we set the current source to zero, as shown:



Applying KVL to the loop gives

$$12i_1 - 6 = 0 \Rightarrow i_1 = 0.5A$$

Thus,

$$v_1 = 4i_1 = 2V$$

We may also use voltage division to get v_1 by writing

$$v_1 = \frac{4}{4+8}(6) = 2V$$

To get v_2 , we set the voltage source to zero, as shown:

Using current division,

$$i_3 = \frac{8}{4+8}(3) = 2A$$

Hence,

$$v_2 = 4i_3 = 8V$$

And we find

$$v = v_1 + v_2 = 2 + 8 = 10V$$

Source Transformation

- A *source transformation* is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa. Source transformation is another tool for simplifying circuits.

Independent sources

- Recall: An equivalent circuit is one whose v - i characteristics are identical with the original circuit.

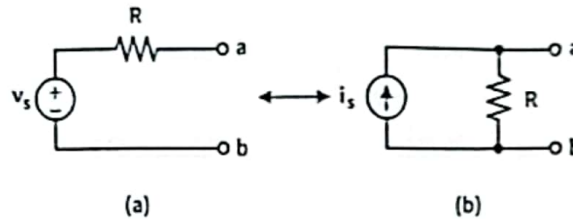


Figure 1: Transformation of independent sources; the two circuits are equivalent—provided they have the same v - i relation at terminals a - b .

- If the sources are turned off, the resistance at terminals a - b in both circuits is R . When terminals a - b are short-circuited, the short-circuit current from a to b is $i_{sc} = v_s/R$ in circuit (a) and $i_{sc} = i_s$ for circuit (b). Thus, $v_s/R = i_s$ in order for the two circuits to be equivalent.
- Hence, source transformation requires that: $v_s = i_s R$ or $i_s = \frac{v_s}{R}$

Dependent sources

- Source transformation also applies to dependent sources, by carefully handling the dependent variable.

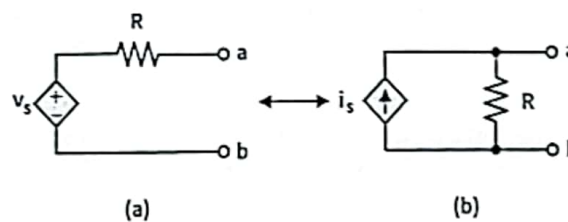


Figure 2: Transformation of dependent sources; a dependent voltage source in series with a resistor transformed to a dependent current source in parallel with the resistor/vice versa where the v - i relation is satisfied.

- Keep the following points in mind when dealing with source transformation.
 - Note from the circuits that the arrow of the current source is directed toward the positive terminal of the voltage source.
 - From the v_s - i_s relation, source transformation is not possible when $R=0$ (ideal voltage source). For a non-ideal voltage source, R is not equal to zero. Similarly, an ideal current source with infinite R cannot be replaced by a finite voltage source.
- Source transformation does not affect the remaining part of the circuit. When applicable, source transformation

eases circuit analysis.

Thevenin's Theorem

- Thevenin's theorem: a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.
- The Thevenin equivalent circuit was developed in 1883 by M. Leon Thevenin (1857–1926), a French telegraph engineer.
- The fixed part of the circuit is replaced by an equivalent circuit to avoid analyzing the circuit all over again when the variable elements change.
- The linear circuit Fig.3(a) can be replaced by Fig.3(b). The load may be a single resistor or another circuit.

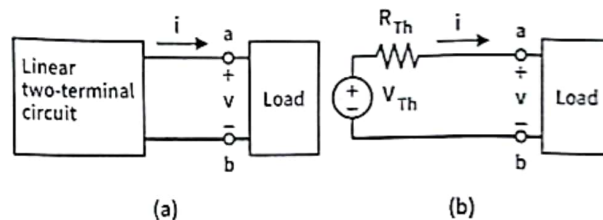


Figure 3: Replacing a linear two-terminal circuit by its Thevenin equivalent: (a) original circuit, (b) the Thevenin equivalent circuit.

Finding V_{Th} and R_{Th} : suppose two circuits are equivalent; that is, they have the same v - i relation at their terminals.

- To find the Thevenin equivalent voltage V_{Th} : if the terminals a-b are made open-circuited, no current flows, so that the open-circuit voltage across the terminals a-b in Fig.3(a) must be equal to the voltage source V_{Th} in Fig.3(b), since the two circuits are equivalent.

$$V_{Th} = v_{oc}$$

- To find R_{Th} : with the load disconnected and a-b open-circuited, we turn off all independent sources. The input or equivalent resistance of the dead circuit at the terminals a-b in Fig.3(a) must be equal to R_{Th} in Fig.3(b) because the two circuits are equivalent.

$$R_{Th} = R_{in}$$

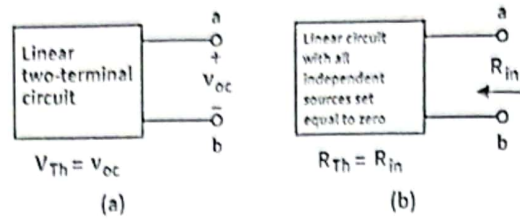


Figure 4: (a) the open-circuit voltage across the terminals while (b) the input resistance at the terminals when the independent sources are turned off.

To apply this idea in finding R_{Th} , there are two cases to consider.

CASE 1: If the network has no dependent sources, we turn off all independent sources. R_{Th} is the input resistance of the network looking between terminals a and b, as shown in Fig. 4(b).

CASE 2: If the network has dependent sources, turn off all independent sources. Dependent sources are not to be turned off as they are controlled by circuit variables. Apply a voltage source v_o at terminals a and b and determine the resulting current i_o . Then, $R_{Th} = v_o / i_o$, as shown in Fig. 5(a). Alternatively, we may insert a current source i_o at a-b as shown in Fig. 5(b) and find the terminal voltage v_o . Again, $R_{Th} = v_o / i_o$. Either of the two approaches will give the same result. We may assume any value of v_o and i_o .

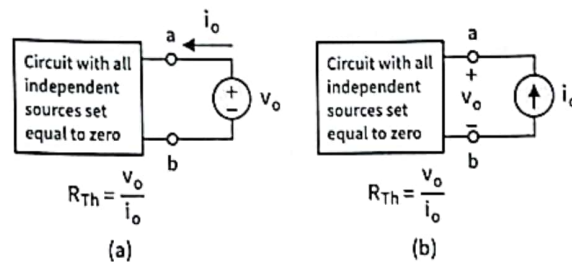


Figure 5: Finding the resistance when the circuit has dependent sources.

R_{Th} can take a negative value. The negative resistance implies that the circuit is supplying power. This is possible in a circuit with dependent sources.

Consider a linear circuit terminated by a load R_L , as shown in Fig. 6(a). The current I_L through the load and the voltage V_L across the load are easily determined once the Thevenin equivalent of the circuit at the load's terminals is obtained as shown in Fig. 6(b).

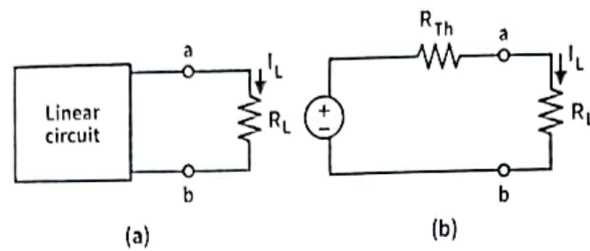


Figure 6: (a) Original circuit with load, (b) Thevenin equivalent.

From Fig.6(b), we obtain:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

Note from Fig.6(b) that the Thevenin equivalent is a simple voltage divider, yielding V_L by mere inspection.

Norton's Theorem

- Norton's theorem: a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.
- Thus, the circuit in Fig.7(a) can be replaced by the one in Fig.7(b).

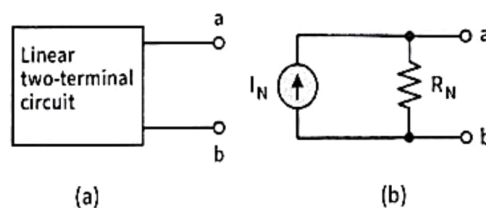


Figure 7: (a) Original circuit, (b) Norton equivalent circuit.

- E. L. Norton, an American engineer at Bell Telephone Laboratories, proposed the theorem in 1926.
- The main concern is to get R_N and I_N .
- We find R_N in the same way we find R_{Th} . In fact, from what we know about source transformation, the Thevenin and Norton resistances are equal; that is,

$$R_N = R_{Th}$$

- To find the Norton current I_N , determine the short-circuit current flowing from terminal a to b in both circuits in Fig. 7. It is evident that the short-circuit current in Fig. 7(b) is I_N . This must be the same short-circuit current from terminal a to b in Fig. 7(a), since the two circuits are equivalent. Thus,

$$I_N = i_{sc}$$

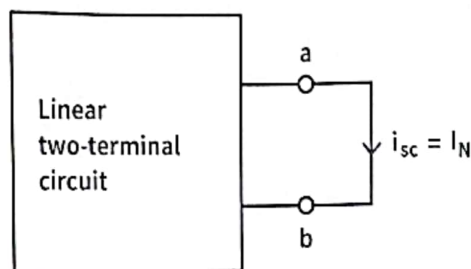


Figure 8: Finding Norton current

- Dependent and independent sources are treated the same way as in Thevenin's theorem.
- Observe the close relationship between Norton's and Thevenin's theorems: $R_N = R_{Th}$, and

$$I_N = \frac{V_{Th}}{R_{Th}}$$

- This is essentially source transformation. For this reason, source transformation is often called Thevenin-Norton transformation.
- Since V_{Th} , I_N , and R_{Th} are related, to determine the Thevenin or Norton equivalent circuit requires that we find:
 1. The open-circuit voltage v_{oc} across terminals a and b .
 2. The short-circuit current i_{sc} at terminals a and b .
 3. The equivalent or input resistance R_{in} at terminals a and b when all independent sources are turned off.

- We can calculate any two of the three using the method that takes the least effort and use them to get the third using Ohm's law. Also, since the open-circuit and short-circuit tests are sufficient to find any Thevenin or Norton equivalent of a circuit which contains at least one independent source.

$$V_{Th} = v_{oc}$$

$$I_N = i_{sc}$$

$$R_{Th} = \frac{v_{oc}}{i_{sc}} = R_N$$

Maximum Power Transfer

- *Maximum power* is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).
- The Thevenin equivalent is useful in finding the maximum power a linear circuit can deliver to a load. We assume that we can adjust the load resistance R_L . If the entire circuit is replaced by its Thevenin equivalent except for the load, as shown in Fig.9, the power delivered to the load is

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad (\text{power equation})$$

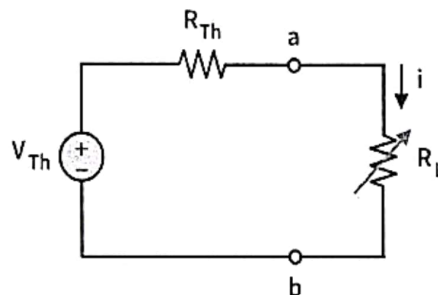


Figure 9: The circuit used for maximum power transfer.

- For a given circuit, V_{Th} and R_{Th} are fixed. By varying R_L , the power delivered to the load varies as sketched in Fig.10.

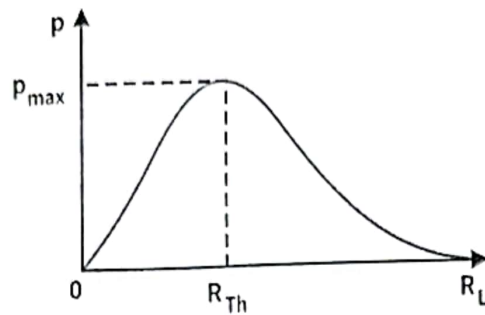


Figure 10: Power delivered to the load.

- The power is small for small or large values of R_L but maximum for some value of R_L between 0 and ∞ .
- Differentiating the power equation and setting the result equal to zero confirms that the maximum power occurs when R_L is equal to R_{Th} . This is known as the *maximum power theorem*: $R_L = R_{Th}$.
- Substituting $R_L = R_{Th}$ to the power equation gives the maximum power transferred

$$p_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

- The equation above applies only when $R_L = R_{Th}$. When R_L is not equal to R_{Th} , we compute the power delivered to the load using the power equation (as stated above).